

Milestone 1

Group Members-

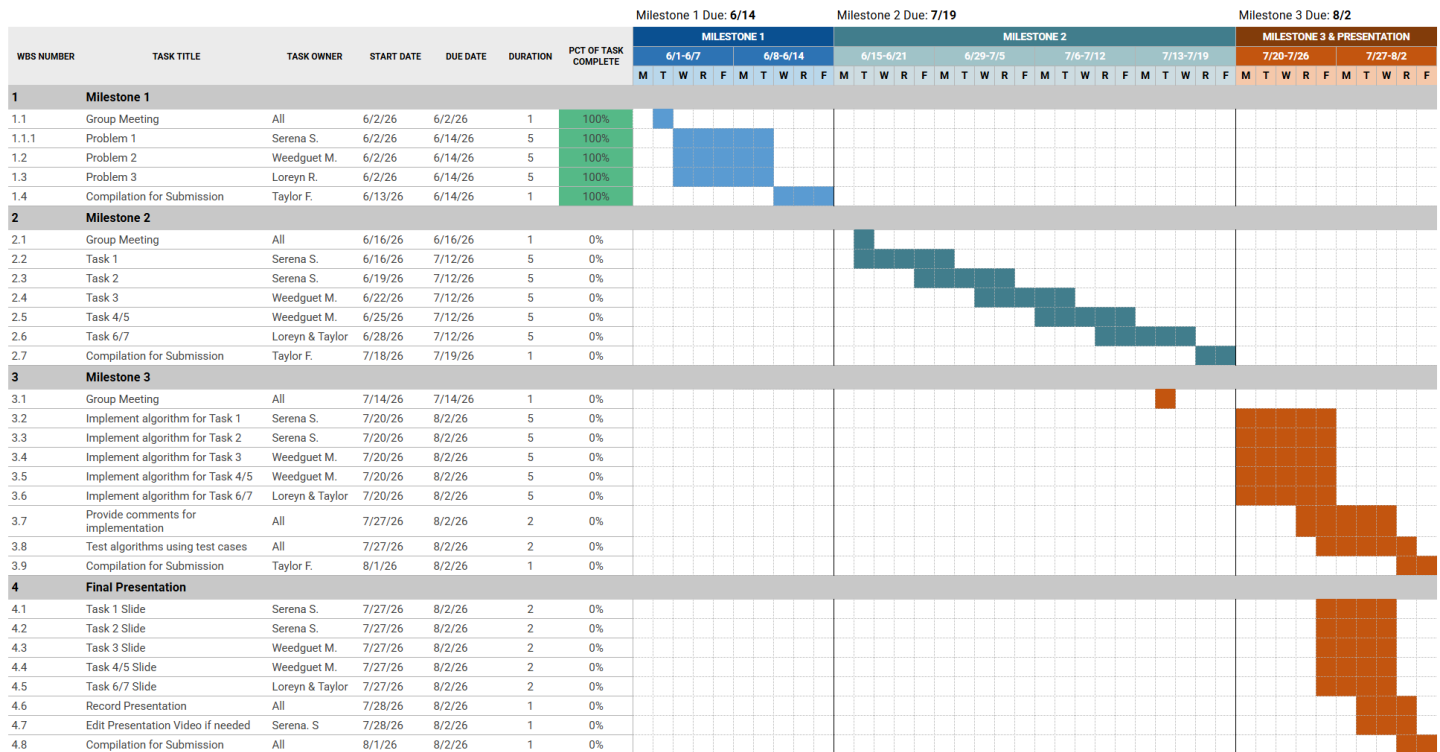
- Taylor Filson (tyler.filson@ufl.edu, GitHub: TFilson)
- Weedguet Mildort (weedguet.mildort@ufl.edu, GitHub: weedguetmildort)
- Loreyn Reid (l.reid1@ufl.edu, GitHub: loreyn626)
- Serena Seymour (serenaseymour@ufl.edu, GitHub: serenaseymour)

Member Roles-

- Taylor Filson: Development, Time Keeper
- Weedguet Mildort: Development
- Loreyn Reid: Development
- Serena Seymour: Development

Communication Method- Discord

Project Gantt Chart-



GitHub Repo- <https://github.com/serenaseymour/COP4533-Group-1-Summer-2026>

Solutions-

Problem Statement 1:

$$A = \begin{matrix} & \begin{matrix} \text{Day 1} & \text{Day 2} & \text{Day 3} & \text{Day 4} & \text{Day 5} \end{matrix} \\ \begin{matrix} \text{Stock 1} \\ \text{Stock 2} \\ \text{Stock 3} \\ \text{Stock 4} \end{matrix} & \begin{bmatrix} 12 & 1 & 5 & 3 & 16 \\ 4 & 4 & 13 & 4 & 9 \\ 6 & 8 & 6 & 1 & 2 \\ 14 & 3 & 4 & 8 & 10 \end{bmatrix} \end{matrix}$$

Stock 1				
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome
1	2	1-12	-11	Loss
1	3	5-12	-7	Loss
1	4	3-12	-9	Loss
1	5	16-12	4	Good
2	3	5-1	4	Good
2	4	3-1	2	Good
2	5	16-1	15	Best
3	4	3-5	-2	Loss
3	5	16-5	11	Good
4	5	16-3	13	Good

Stock 2				
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome
1	2	4-4	0	Even
1	3	13-4	9	Best
1	4	4-4	0	Even
1	5	9-4	5	Good
2	3	13-4	9	Best
2	4	4-4	0	Even
2	5	9-4	5	Good
3	4	13-4	9	Best
3	5	13-9	5	Good
4	5	4-9	-5	Loss

Stock 3				
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome
1	2	8-6	2	Best
1	3	6-6	0	Even
1	4	1-6	-5	Loss
1	5	2-6	-4	Loss
2	3	6-8	-2	Loss
2	4	1-8	-7	Loss
2	5	2-8	-6	Loss
3	4	1-6	-5	Loss
3	5	2-6	-4	Loss
4	5	2-1	-1	Loss

Stock 4				
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome
1	2	3-14	-11	Loss
1	3	4-14	-10	Loss
1	4	8-14	-6	Loss
1	5	10-14	-4	Loss
2	3	4-3	1	Good
2	4	8-3	5	Good
2	5	10-3	7	Best
3	4	8-4	4	Good
3	5	10-4	6	Good
4	5	10-8	2	Good

Stock 1 Best: **15**

Stock 2 Best: 9

Stock 3 Best: 2

Stock 4 Best: 7

Max Profit= 15
(1, 2, 5, 15)

Stock 1, bought on day 2 (1), sold on day 5 (16), profit= 15

Problem Statement 2:

$$A = \begin{matrix} & \begin{matrix} \text{Day 1} & \text{Day 2} & \text{Day 3} & \text{Day 4} & \text{Day 5} \end{matrix} \\ \begin{matrix} \text{Stock 1} \\ \text{Stock 2} \\ \text{Stock 3} \\ \text{Stock 4} \end{matrix} & \begin{bmatrix} 25 & 30 & 15 & 40 & 50 \\ 10 & 20 & 30 & 25 & 5 \\ 30 & 45 & 35 & 10 & 15 \\ 5 & 50 & 35 & 25 & 45 \end{bmatrix} \end{matrix}$$

$$k = 3$$

Stock 1					Stock 2				
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome	Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome
1	2	30-25	5	Gain	1	2	20-10	10	Gain
1	3	15-25	-10	Loss	1	3	30-10	20	Best
1	4	40-25	15	Gain	1	4	25-10	15	Gain
1	5	50-25	25	Gain	1	5	5-10	-5	Loss
2	3	15-30	-15	Loss	2	3	30-20	10	Gain
2	4	40-30	10	Gain	2	4	25-20	5	Gain
2	5	50-30	20	Gain	2	5	5-20	-15	Loss
3	4	40-15	25	Gain	3	4	25-30	-5	Loss
3	5	50-15	35	Best	3	5	5-30	-25	Loss
4	5	50-40	10	Gain	4	5	5-25	-20	Loss

Stock 3					Stock 4				
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome	Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome
1	2	45-30	15	Best	1	2	50-5	45	Best
1	3	35-30	5	Gain	1	3	35-5	30	Gain
1	4	10-30	-20	Loss	1	4	25-5	20	Gain
1	5	15-30	-15	Loss	1	5	45-5	40	Gain
2	3	35-45	-10	Loss	2	3	35-50	-15	Loss
2	4	10-45	-35	Loss	2	4	25-50	-25	Loss
2	5	15-45	-30	Loss	2	5	45-50	-5	Loss
3	4	10-35	-25	Loss	3	4	25-35	-10	Loss
3	5	15-35	-20	Loss	3	5	45-35	10	Gain
4	5	15-10	5	Gain	4	5	45-25	20	Gain

All the best profits:

Stock 1 Best: 35

Stock 2 Best: 20

Stock 3 Best: 15

Stock 4 Best: 45

The top two best profits are 35 and 45, and they are acquired between day 1 to day 2, and between day 3 to day 5, which leaves day 2 and day 3 available to take another transaction.

All the results between day 2 and 3:

Stock 1 Best: -15

Stock 2 Best: 10

Stock 3 Best: -10

Stock 4 Best: -15

The best result between day 2 and day 3 is stock 2.

A sequence of at most k transactions, each involving the purchase and sale of a single stock, that yields the maximum profit is: [(4, 1, 2),(2, 2, 3),(1, 3, 5)]

Stock 4, bought on day 1 (5), sold on day 2 (50), profit= 45

Stock 2, bought on day 2 (20), sold on day 3 (30), profit= 10

Stock 1, bought on day 3 (15), sold on day 5 (50), profit= 35

Hence the total maximum possible profit is 90. To reach the maximum possible profit, the first step was to identify the two best transactions, then determine the best third transaction that satisfied all constraints.

Problem Statement 3:

$$A = \begin{matrix} & \begin{matrix} \text{Day 1} & \text{Day 2} & \text{Day 3} & \text{Day 4} & \text{Day 5} & \text{Day 6} & \text{Day 7} \end{matrix} \\ \begin{matrix} \text{Stock 1} \\ \text{Stock 2} \\ \text{Stock 3} \\ \text{Stock 4} \\ \text{Stock 5} \end{matrix} & \begin{bmatrix} 7 & 1 & 5 & 3 & 6 & 8 & 9 \\ 2 & 4 & 3 & 7 & 9 & 1 & 8 \\ 5 & 8 & 9 & 1 & 2 & 3 & 10 \\ 9 & 3 & 4 & 8 & 7 & 4 & 1 \\ 3 & 1 & 5 & 8 & 9 & 6 & 4 \end{bmatrix} \end{matrix} \quad c = 2$$

Based on our current example of 7 consecutive days and the 2 day trading restriction, the max number of full transactions (buy and sell) is 2. To be able to make a second full transaction, the sell day of transaction 1 can be no later than day 3. The purchase date of transaction 2 can be no later than day 6.

T1: Transaction 1 Results

Note: Eligibility to buy again is based on the statement above. If the sell day of transaction 1 is after day 3, then for calculation purposes, it is marked not eligible to buy again. There would not be enough time to make a second full transaction given the trading restriction.

Best E refers to the highest profit of those eligible to buy again. Best O is the overall highest profit.

Stock 1					
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome	Eligible to Buy Again (E)
1	2	1-2	-6	Loss	Y - On Days 5 or 6
1	3	2-3	-2	Loss	Y - On Day 6
1	4	3-4	-4	Loss	N
1	5	4-5	-1	Loss	N
1	6	5-6	1	Gain	N
1	7	6-7	2	Gain	N
2	3	1-2	4	Gain (Best E)	Y - On Day 6
2	4	2-3	2	Gain	N
2	5	3-4	5	Gain	N
2	6	4-5	7	Gain	N
2	7	5-6	8	Gain (Best O)	N
3	4	1-2	-2	Loss	N
3	5	2-3	1	Gain	N
3	6	3-4	3	Gain	N
3	7	4-5	4	Gain	N
4	5	1-2	3	Gain	N
4	6	2-3	5	Gain	N
4	7	3-4	6	Gain	N
5	6	1-2	2	Gain	N
5	7	2-3	3	Gain	N
6	7	1-2	1	Gain	N

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Stock 2					
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome	Eligible to Buy Again (E)
1	2	4-2	2	Gain (Best E)	Y - On Days 5 or 6
1	3	3-2	1	Gain	Y - On Day 6
1	4	7-2	5	Gain	N
1	5	9-2	7	Gain (Best O)	N
1	6	1-2	-1	Loss	N
1	7	8-2	6	Gain	N
2	3	3-4	-1	Loss	Y - On Day 6
2	4	7-4	3	Gain	N
2	5	9-4	5	Gain	N
2	6	1-4	-3	Loss	N
2	7	8-4	4	Gain	N
3	4	7-3	4	Gain	N
3	5	9-3	6	Gain	N
3	6	1-3	-2	Loss	N
3	7	8-3	5	Gain	N
4	5	9-7	2	Gain	N
4	6	1-7	-6	Loss	N
4	7	8-7	1	Gain	N
5	6	1-9	-8	Loss	N
5	7	8-9	-1	Loss	N
6	7	8-1	7	Gain (Best O)	N

Stock 3					
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome	Eligible to Buy Again (E)
1	2	8-5	3	Gain	Y - On Days 5 or 6
1	3	9-5	4	Gain (Best E)	Y - On Day 6
1	4	1-5	-4	Loss	N
1	5	2-5	-3	Loss	N
1	6	3-5	-2	Loss	N
1	7	10-5	5	Gain	N
2	3	9-8	1	Gain	Y - On Day 6
2	4	1-8	-7	Loss	N
2	5	2-8	-6	Loss	N
2	6	3-8	-5	Loss	N
2	7	10-8	2	Gain	N
3	4	1-9	-8	Loss	N
3	5	2-9	-7	Loss	N
3	6	3-9	-6	Loss	N
3	7	10-9	1	Gain	N
4	5	2-1	1	Gain	N
4	6	3-1	2	Gain	N
4	7	10-1	9	Gain (Best O)	N
5	6	3-2	1	Gain	N
5	7	10-2	8	Gain	N
6	7	10-3	7	Gain	N

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Stock 4					
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome	Eligible to Buy Again (E)
1	2	3-9	-6	Loss	Y - On Days 5 or 6
1	3	4-9	-5	Loss	Y - On Day 6
1	4	8-9	-1	Loss	N
1	5	7-9	-2	Loss	N
1	6	4-9	-5	Loss	N
1	7	1-9	-8	Loss	N
2	3	4-3	1	Gain (Best E)	Y - On Day 6
2	4	8-3	5	Gain (Best O)	N
2	5	7-3	4	Gain	N
2	6	4-3	1	Gain	N
2	7	1-3	-2	Loss	N
3	4	8-4	4	Gain	N
3	5	7-4	3	Gain	N
3	6	4-4	0	Gain	N
3	7	1-4	-3	Loss	N
4	5	7-8	-1	Loss	N
4	6	4-8	-4	Loss	N
4	7	1-8	-7	Loss	N
5	6	4-7	-3	Loss	N
5	7	1-7	-6	Loss	N
6	7	1-4	-3	Loss	N

Stock 5					
Buy Day (j1)	Sell Day (j2)	(j2)-(j1)	Profit	Outcome	Eligible to Buy Again (E)
1	2	1-3	-2	Loss	Y - On Days 5 or 6
1	3	5-3	2	Gain	Y - On Day 6
1	4	8-3	5	Gain	N
1	5	9-3	6	Gain	N
1	6	6-3	3	Gain	N
1	7	4-3	1	Gain	N
2	3	5-1	4	Gain (Best E)	Y - On Day 6
2	4	8-1	7	Gain	N
2	5	9-1	8	Gain (Best O)	N
2	6	6-1	5	Gain	N
2	7	4-1	3	Gain	N
3	4	8-5	3	Gain	N
3	5	9-5	4	Gain	N
3	6	6-5	1	Gain	N
3	7	4-5	-1	Loss	N
4	5	9-8	1	Gain	N
4	6	6-8	-2	Loss	N
4	7	4-8	-4	Loss	N
5	6	6-9	-3	Loss	N
5	7	4-9	-5	Loss	N
6	7	4-6	-2	Loss	N

T2: Transaction 2 Results & Final Profits

For each transaction that was eligible for a second transaction, we compute the total profit of each combination of T1 and T2 to find the max profit.

Note: The Max Profit takes into account single transactions from the results above and the total profits from the 2nd transaction results below.

Transaction 2 (Stock 1 Transaction 1)						
Stock	Buy Day (j1)	Sell Day (j2)	Profit	Total Profit, T1: j1 = 1, j2 = 2, (-6 + profit)	Total Profit T1: j1 = 1, j2 = 3, (-2 + profit)	Total Profit T1: j1 = 2, j3 = 3, (4 + profit)
1	5	6	2	-4		
1	5	7	3	-3		
1	6	7	1	-5	-1	5
2	5	6	-8	-14		
2	5	7	-1	-7		
2	6	7	7	1	5	11
3	5	6	1	-5		
3	5	7	8	2		
3	6	7	7	1	5	11
4	5	6	-3	-9		
4	5	7	-6	-12		
4	6	7	-3	-9	-5	1
5	5	6	-3	-9		
5	5	7	-5	-11		
5	6	7	-2	-8	-4	2

Stock 1:

Max Profit = 11

[(1, 2, 3), (2,6,7)]

[(1, 2, 3), (3,6,7)]

Transaction 2 (Stock 2 Transaction 1)						
Stock	Buy Day (j1)	Sell Day (j2)	Profit	Total Profit, T1: j1 = 1, j2 = 2, (2 + profit)	Total Profit T1: j1 = 1, j2 = 3, (1 + profit)	Total Profit T1: j1 = 2, j3 = 3, (-1 + profit)
1	5	6	2	4		
1	5	7	3	5		
1	6	7	1	3	2	0
2	5	6	-8	-6		
2	5	7	-1	1		
2	6	7	7	9	8	6
3	5	6	1	3		
3	5	7	8	10		
3	6	7	7	9	8	6
4	5	6	-3	-1		
4	5	7	-6	-4		
4	6	7	-3	-1	-2	-4
5	5	6	-3	-1		
5	5	7	-5	-3		
5	6	7	-2	0	-1	-3

Stock 2:

Max Profit = 10

[(2, 1, 2), (3,5,7)]

Transaction 2 (Stock 3 Transaction 1)						
Stock	Buy Day (j1)	Sell Day (j2)	Profit	Total Profit, T1: j1 = 1, j2 = 2, (3 + profit)	Total Profit T1: j1 = 1, j2 = 3, (4 + profit)	Total Profit T1: j1 = 2, j3 = 3, (1 + profit)
1	5	6	2	5		
1	5	7	3	6		
1	6	7	1	4	5	2
2	5	6	-8	-5		
2	5	7	-1	2		
2	6	7	7	10	11	8
3	5	6	1	4		
3	5	7	8	11		
3	6	7	7	10	11	8
4	5	6	-3	0		
4	5	7	-6	-3		
4	6	7	-3	0	1	-2
5	5	6	-3	0		
5	5	7	-5	-2		
5	6	7	-2	1	2	-1

Stock 3:

Max Profit = 11

[(3, 1, 2), (3,5,7)]

[(3, 1, 3), (2,6,7)]

[(1, 1, 3), (3,6,7)]

Transaction 2 (Stock 4 Transaction 1)						
Stock	Buy Day (j1)	Sell Day (j2)	Profit	Total Profit, T1: j1 = 1, j2 = 2, (-6 + profit)	Total Profit T1: j1 = 1, j2 = 3, (-5 + profit)	Total Profit T1: j1 = 2, j3 = 3, (1 + profit)
1	5	6	2	-4		
1	5	7	3	-3		
1	6	7	1	-5	-4	2
2	5	6	-8	-14		
2	5	7	-1	-7		
2	6	7	7	1	2	8
3	5	6	1	-5		
3	5	7	8	2		
3	6	7	7	1	2	8
4	5	6	-3	-9		
4	5	7	-6	-12		
4	6	7	-3	-9	-8	-2
5	5	6	-3	-9		
5	5	7	-5	-11		
5	6	7	-2	-8	-7	-1

Stock 4:

Max Profit = 8

[(4, 2, 3), (2,6,7)]

[(4, 2, 3), (3,6,7)]

Transaction 2 (Stock 5 Transaction 1)						
Stock	Buy Day (j1)	Sell Day (j2)	Profit	Total Profit, T1: j1 = 1, j2 = 2, (-2 + profit)	Total Profit T1: j1 = 1, j2 = 3, (2 + profit)	Total Profit T1: j1 = 2, j3 = 3, (4 + profit)
1	5	6	2	0		
1	5	7	3	1		
1	6	7	1	-1	3	5
2	5	6	-8	-10		
2	5	7	-1	-3		
2	6	7	7	5	9	11
3	5	6	1	-1		
3	5	7	8	6		
3	6	7	7	5	9	11
4	5	6	-3	-5		
4	5	7	-6	-8		
4	6	7	-3	-5	-1	1
5	5	6	-3	-5		
5	5	7	-5	-7		
5	6	7	-2	-4	0	2

Stock 5:

Max Profit = 11

[(5, 2, 3), (2,6,7)]

[(5, 2, 3), (3,6,7)]

The overall max profit is 11. This could be obtained through multiple transactions. All are listed below.

[(1, 2, 3), (2,6,7)]

[(1, 2, 3), (3,6,7)]

[(3, 1, 2), (3,5,7)]

[(3, 1, 3), (2,6,7)]

[(1, 1, 3), (3,6,7)]

[(5, 2, 3), (2,6,7)]

[(5, 2, 3), (3,6,7)]

We only need to output one of these results:

Output: [(1, 2, 3), (2,6,7)]